Arda Fazla

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Education

Doctor of Philosophy **- Electrical and Computer Engineering** 2024-Present
Purdue University - The Elmore Family School of Electrical and Computer Engineering Indiana, US
Advisor: Prof. Abolfazl Hashemi

Master of Science **- Electrical and Electronics Engineering** 2021-2024
Bilkent University - Faculty of Engineering Ankara, Turkey
Advisor: Prof. Süleyman Serdar Kozat

Bachelor of Science **- Electrical and Electronics Engineering** 2017-2021
Middle East Technical University - Faculty of Engineering Ankara, Turkey

Research Background & Interests

- **Optimization Theory:** Convergence analysis, variance-reduction methods
- **LLM Training:** Efficient large-scale training, adaptive and spectral optimizers
- **Reinforcement Learning:** RL-based post-training, LLM alignment
- **Trustworthy Deep Learning:** Robustness, fairness, memorization

Professional Experiences

Purdue University, *Research Assistant (Advisor: Prof. Abolfazl Hashemi)* West Lafayette, IN

• *Stochastic Optimization* Aug 2025 - Present

- Developed PASTA and SAUCE, two stochastic optimization frameworks for nonconvex learning under realistic noise assumptions, covering the unbounded variance under the Blum-Gladyshev noise. Established theoretical convergence guarantees and demonstrated their relevance to modern deep learning settings where classical bounded-variance assumptions fail.
- Developed an adaptive batch size scheduler for LLM pretraining, motivated by the observation that gradient noise grows with distance from initialization (unbounded variance). Provided convergence guarantees under this noise model and showed lower training and validation loss than fixed-batch baselines at matched compute.

• *Trustworthy Deep Learning* Aug 2024 - Present

- Proposed a two-stage coreset selection algorithm with the dual purpose of eliminating spurious correlations and reducing dataset size. Deployed product achieved up to **27%** improvement over standard ERM utilizing only **10%** of the training data on benchmark datasets such as Waterbirds, CMNIST, MetaShift and UrbanCars, while also outperforming state-of-the-art invariant learning methods.
- Currently studying the training dynamics of deep learning models, with a focus on how loss landscape sharpness interacts with common training interventions and influences memorization and generalization, particularly in out-of-distribution settings.

Databoss Security & Analytics Inc., *Data Scientist* Ankara, Turkey

• *Hourly Wind Energy Prediction* Jun 2023 - May 2024

- Predicted the hourly energy data of multiple wind turbines in Turkey and Europe, showing chaotic and nonstationary behavior in multiple regions.
- Developed a large-scale ML framework consisting of various adaptive feature construction and selection methods with models such as deep-learning (NN, ConvLSTM), machine learning (LightGBM, RandomForest), statistical (SARIMAX) models and state-of-the-art ensembling techniques. The deployed product increased the short-term forecasting accuracy by an average of **21%** and long-term forecasting accuracy by an average of **8%**.

• *Natural Gas Demand Prediction* Aug 2021 - May 2023

- Predicted the daily data of international natural gas demand in Turkey, consisting of nonstationary multivariate time series from multiple sources.
- Developed a large-scale ML framework specifically tailored towards a certain customer profile, where the predictions are obtained for the given data based on key indicators that are automatically extracted from the data itself and/or the user-given side information data. The constructed framework was deployed to various natural gas production companies.

METU Center for Image Analysis (OGAM), *Machine Learning Engineer* Ankara, Turkey

• *UAV Small Object Detection and Tracking* Jun 2020 - Jul 2021

- Developed a Python application for real-time small target tracking by leveraging temporal data derived from the output of object detectors, with a superior performance over the widely-known YOLOv3 algorithm by **6%**.
- Constructed a novel multi-hypothesis tracking algorithm with an annotation tool for the semi-automatic correction of mislabeled UAV data. The deployed product decreased the total mislabeled data in various datasets by an average of **13%**.

Publications

Preprints

[1] M. Sahin, E. Kaya, **A. Fazla**, B. Sharif, A. Hashemi, *"Robust Langevin Sampling under State-Dependent Heavy-Tailed Noise with Applications to LLM Fine-Tuning"*, Under Review at AISTATS 2027.

[2] **A. Fazla**, A. Upadhyay, E. Kaya, M. Sahin, A. Hashemi, *"Why Does Adaptive Batching Help LLM Pretraining? A Perspective from Unbounded Variance"*, Under Review at ICLR 2027.

[3] E. Kaya, **A. Fazla**, M. Sahin, A. Hashemi, *"A Banach-Space Theory of Markovian Halpern Iteration for Non-Expansive Maps"*, Under Review at ICLR 2027.

[4] **A. Fazla**, E. Kaya, A. Upadhyay, A. Hashemi, *"Lower Bounds and Proximally Anchored SGD for Non-Convex Minimization Under Unbounded Variance"*, Under Review at ICLR 2027.

Published

[5] A. Upadhyay, **A. Fazla**, A. Hashemi, *"Beyond Bounded Variance: Variance-Reduced Normalized Methods for Nonconvex Optimization under Blum-Gladyshev Noise"*, Neural Information Processing Systems (NeurIPS) 2026.

[6] **A. Fazla**, A. Hashemi, *"Mitigating Spurious Correlations with Memorization-Guided Dataset De-Biasing"*, Uncertainty in Artificial Intelligence (UAI), 2026.

[7] S. F. Tekin, **A. Fazla**, S. S. Kozat, *"Numerical Weather Forecasting using Convolutional-LSTM with Attention and Context Matcher Mechanisms"*, IEEE Transactions on Geoscience and Remote Sensing, 2024.

[8] **A. Fazla**, M. E. Aydin, S. S. Kozat, *"Time-Aware and Context-Sensitive Ensemble Learning for Sequential Data"*, IEEE Transactions on Artificial Intelligence, 2023.

[9] M. E. Aydin, **A. Fazla**, S. S. Kozat, *"Hybrid State Space-based Learning for Sequential Data Prediction with Joint Optimization"*, arXiv:2309.10553, 2023.

[10] **A. Fazla**, M. E. Aydin, S. S. Kozat, *"Joint Optimization of Linear and Nonlinear Models for Sequential Regression"*, Digital Signal Processing, Elsevier, 2022.

[11] K. G. Ince, A. Köksal, **A. Fazla**, A. A. Alatan, *"Semi-Automatic Annotation for Object Detection"*, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2021, pp. 1233-1239.

Academic Awards

Elmore Family School of Electrical and Computer Engineering, Purdue University 2024-2029

• Awarded the highly competitive and prestigious Elmore Fellowship Award for my graduate studies, given only to the most outstanding applicants.

Turk Telekom & Information and Communication Technologies Authority 2023-2024

• Awarded 5G and Beyond Joint Graduate Support Programme, a merit-based fellowship of monthly stipend during M.Sc.

Scientific and Technological Research Council of Turkey 2021-2023

• Awarded Directorate of Science Fellowships and Grant Programme, a merit-based monthly stipend and accommodation support during M.Sc.

• Received the **373rd** rank among 2M high school graduates in the University Entrance Examination.

• Received the **11th** rank among 0.2M university graduates in ALES (National GRE).

Academic Duties

Reviewer Duties

• **Conferences:** ICLR '25-'27, NIPS '26, AISTATS '26-'27, AAAI '25
• **Journals:** IEEE TAI '25, IEEE IOT '24-'25

Teaching Assistantship at Bilkent University, Electrical and Electronics Engineering 2021-2024

• EEE321 Signals and Systems, EEE202 Circuit Theory, MATH255 Probability and Statistics

Software Skills

- Python: Professional research and industrial experience based on machine learning. High knowledge and experience in libraries such as *PyTorch*, *Tensorflow*, *Pandas*, *Numpy* and *Scikit-Learn*. I have experience with Docker and Git via industrial projects.
- MATLAB: Professional research experience in signal processing and computer vision. Used for projects during courses, research and internships.
- R: Intermediate level experience, employed during research based on time-series forecasting using statistical models.
- C/C++: Intermediate experience in various course projects.

Languages

Turkish: Native
English: Conversationally Fluent [TOEFL IBT 108/120 (October 19, 2022)]